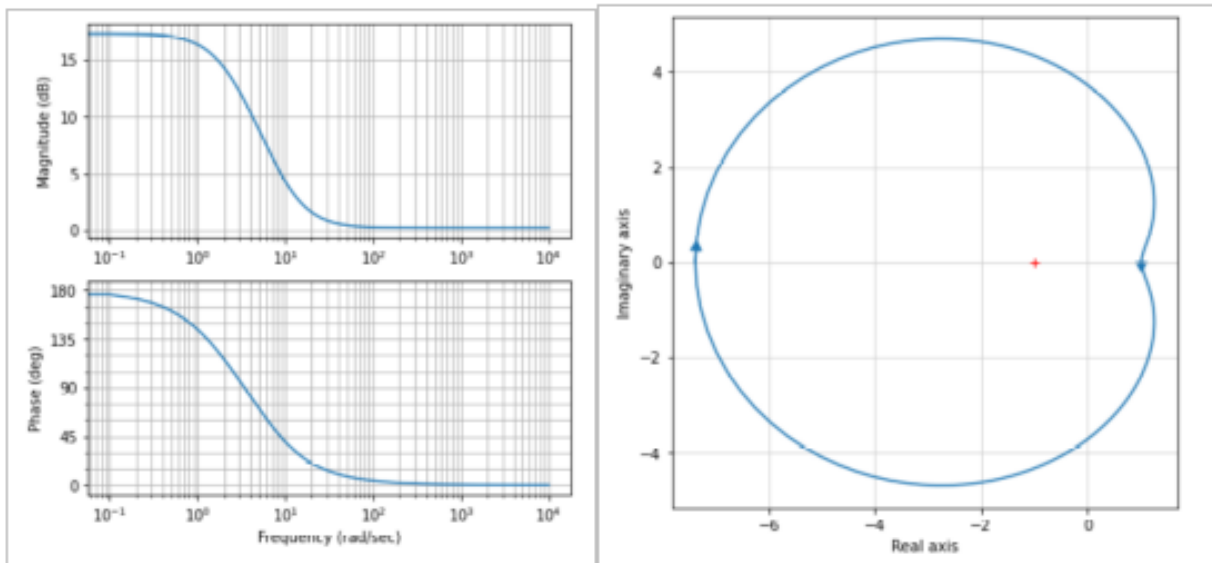


University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering
Problem Set 5.1: Nyquist Plots

Question 1

A given system is unstable in closed-loop.

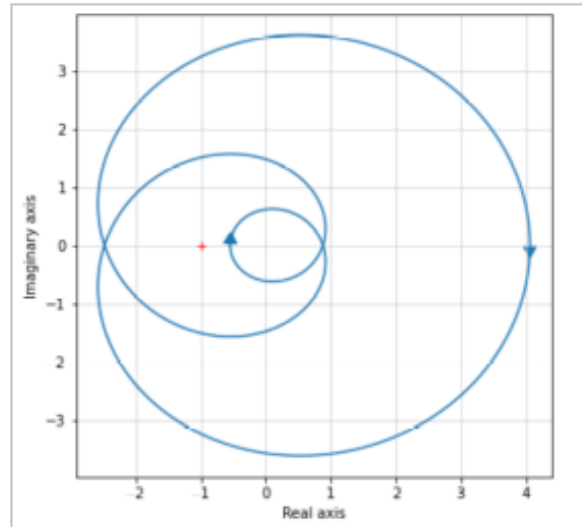
- A) Explain how you would know that from looking at each of the frequency domain plots below
 - i. Bode plot
 - ii. Nyquist plot
- B) How many closed-loop poles are in the right-half plane?
- C) Approximate how much you would have to reduce the gain by to stabilise the system.



Question 2

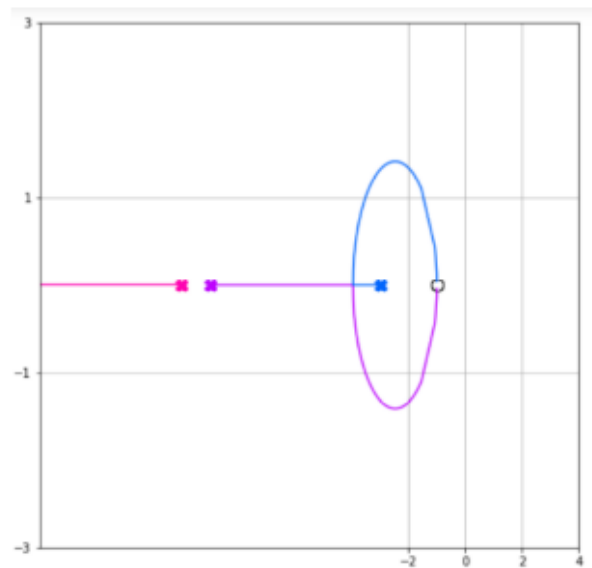
The following Nyquist plot describes a third-order system that is stable in open-loop.

- A) How many unstable closed-loop poles does it have at its present gain?
- B) Will the number of unstable closed-loop poles change if the gain is increased?
- C) How many right-half plane zeros do you expect this system to have based on your previous answers?

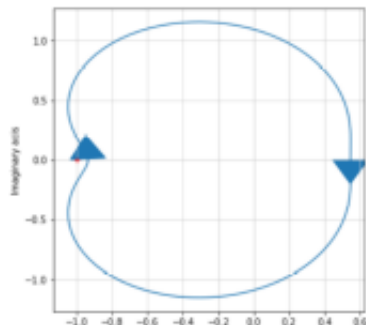


Question 3

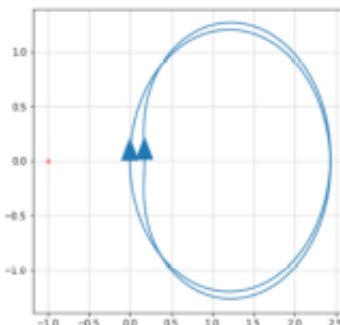
The system associated with the following root locus is closed-loop stable at its present gain.



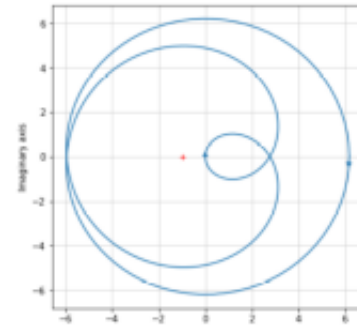
Which of the following Nyquist plots would you say belongs to this system? If a plot does not apply, give a brief reason that you are excluding it.



A



B



C